

Problem 7.9

Find the present (discounted) value of \$3,000 to be paid at the end of 5 years if the accumulation function is

$$a(t) = 1 + \frac{t^2}{25}.$$

Solution

The present value of an amount F due at time t is

$$\text{PV} = \frac{FV}{a(t)}.$$

Here, $F = 3,000$ and $t = 5$, so we first compute

$$a(5) = 1 + \frac{5^2}{25} = 1 + \frac{25}{25} = 2.$$

Therefore,

$$\text{PV} = \frac{3,000}{a(5)} = \frac{3,000}{2} = 1,500.$$

So the present value is

$$\boxed{\$1,500}.$$